

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2028

Bachelor of Science in Computer Science and Biology • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Coursework: Deep Learning, Artificial Intelligence, Advanced Algorithms, Linear Algebra, Differential Equations, Data Structures, Cryptography, Computer Science Capstone, AP Statistics, AP Biology, AP Computer Science A

TECHNOLOGIES

Languages: Python, R, TypeScript, Node.js, Java, JavaScript, C++, SQL, Linux/Bash

Clinical & Health: EHR Integration (HL7/FHIR), EMR Systems, HIPAA, Clinical Validation, Interoperability, Healthcare Data, Health Records, ICD-10, Clinical Decision Support, Patient Risk Scoring, Oncology, Real-World Evidence

Bioinformatics: Scanpy, scVI, scANVI, scGen, AmortizedLDA, scRNA-seq, NGS, Spatial Transcriptomics, Single-Cell Analysis, Gene Expression, Drug Discovery, Precision Medicine, Pipeline Development

ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, NLP, Generative AI, LLM APIs, RAG, Semantic Similarity Search, Random Forest, Statistical Modeling, Cross-Validation

Systems: Kafka, Docker, MongoDB, FastAPI, REST APIs, Microservices, System Design, Cloud, CI/CD, GitHub Actions, Git, Agile, Object-Oriented Design, Scalable Data Pipelines, ETL, pytest, Publication

EXPERIENCE

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Student Research Intern

- Built scalable Python genomics data pipeline for scRNA-seq (NGS) and spatial transcriptomics; **clinical validation** deployed in production healthcare data environment serving oncology/glioblastoma gene therapy at UW-Madison Medical Center; cut GPU costs by **300%**
- Engineered TransferAgent NLP module automating clinical literature synthesis and cross-paper validity scoring; accelerated **drug discovery** and precision medicine iteration by **~60%**; outputs support HL7/FHIR interoperability standards

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built **skin cancer detection pipeline** (OpenCV + scikit-learn/TensorFlow, NumPy/Pandas) achieving **93% classification accuracy**; published as **peer-reviewed paper** through Inspirit AI
- Trained Random Forest on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation

Leidos

May 2025 - Present

Software Engineer Intern

- Designed Kafka event-driven pipeline with **compliance-grade data integrity** and fault-tolerant Docker streaming; deployed on Kubernetes via **GitHub Actions CI/CD**
- Optimized multi-objective satellite scheduling via memoization + DP in Python; maintained **full mission objective alignment** under strict real-time operational constraints
- Architected decoupled, scalable system design for scheduling, pipeline, and monitoring components; maintained data integrity and availability guarantees for downstream operations

TECHNICAL PROJECTS

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built clinical risk scoring engine across **45M insurance records** on **1,080-node ICD-10 graph**; integrates with **HIPAA-compliant, EHR-adjacent** clinical decision support workflows as validated patient-level risk score
- RAG pipeline with generative AI (Groq Llama) matches patient symptoms to ICD-10 disease risk profiles via SQL-queried healthcare data; all clinical outputs **hallucination-free and auditable**
- Full-stack React frontend with Cytoscape.js interactive disease graph and voice interface; object-oriented Python backend enforces deterministic, auditable financial and clinical logic

miRcore Cancer Detection Research

Jul 2021

- Trained and validated Random Forest classifier in R on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation for cancer cell likelihood prediction

Computer Vision Hand Tracking System (<https://github.com/AkashGogate/HandOrientationTracking>)

March 2024

- Built real-time hand gesture + joint angle tracking pipeline (OpenCV + Mediapipe); outputs biomechanical data for clinical motion capture, rehabilitation monitoring, and sports performance analysis

USTA Tournament Explorer (<https://github.com/AkashGogate/myUSTA>)

Jun 2022

- Built Android app in Java consuming USTA's GraphQL API; surfaces tournament listings via REST API — end-to-end mobile development from API design to health-data presentation

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics; coached athletes across skill levels in cross-functional environment; managed end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations